

SIMULATING DIFFUSION OF INFORMATION IN SOCIAL CONTACT NETWORKS

CAPSTONE PROJECT PHASE-1

Phase – I Report

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Bonafide Certificate

Certified that this project report titled **Simulating diffusion of information in social contact networks** is the bonafide work of 21BCE11076 Devashish Dhaulakhandi, 21BCE11072 Aswin P Kumar, 21BCE11078 Braj Kishor Sharma, 21BCE11159 Arijit Modak, 21BCE11615 Akshant Ujjwal who carried out the project work under my supervision.

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Supervisor

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CHAPTER 1: INTRODUCTION

Information diffusion modeling plays a pivotal role in understanding the dynamics of how information spreads across social networks. Our capstone project leverages graph theory and network analysis to simulate this diffusion using tools like Neo4j and NetworkX, along with a simplified SI (Susceptible-Infected) model. By applying these open-source platforms, we are able to visualize the propagation of information and identify the key factors driving its spread. As we advance, we aim to implement agent-based modeling using Mesa to simulate more complex, human-like interactions in networks, thus offering a deeper understanding of social network dynamics.

Information spreads not just through exposure but also through the influence of network structures, content relevance, and individual behavior. Our approach provides a computationally efficient yet robust framework to analyze these dynamics, offering insights into real-world challenges such as misinformation control and viral marketing strategies.

1.1 Motivation

In today's hyper-connected world, the ability to predict and control the spread of information is crucial. Misinformation during public health crises, viral marketing, and political propaganda all demonstrate the real-world consequences of unchecked information flow. Our motivation stems from the need to provide a more accurate, adaptable, and efficient model that can simulate these processes and offer actionable insights. By building on open-source tools like Neo4j and NetworkX and aiming to expand to Mesa for agent-based modeling, we provide a scalable and practical solution to tackle diverse information diffusion challenges.

The future addition of agent-based modeling will enable us to simulate the behavior of individual "agents" within the network, reflecting human-like decision-making and interaction patterns. This will significantly enhance our ability to model complex, decentralized information flows and respond effectively to real-world problems.

1.2 Objective

- **Simulate Information Diffusion using Neo4j and NetworkX:** Utilize Neo4j and NetworkX to model and visualize information flow within social networks using a simplified SI model, identifying patterns in propagation across different nodes.
- **Identify Key Influencers and Critical Nodes:** Analyze network structures to pinpoint key influencers who play a major role in spreading or containing information, providing actionable insights for managing or amplifying diffusion.
- **Analyze Real-World Applications:** Apply the model to practical scenarios such as public health, crisis management, and viral marketing, addressing challenges like misinformation control and campaign optimization.

CHAPTER 2: LITERATURE REVIEW

In the rapidly evolving landscape of digital communication, understanding how information diffuses through social networks is crucial. Compartmental models, inspired by epidemiology, have become instrumental in simulating and analyzing information spread. Among these, the **SI (Susceptible-Infected)** model is particularly noteworthy for its simplicity and effectiveness in capturing the fundamental dynamics of information diffusion.

The SI Model: A Simplified Approach to Information Diffusion

The SI model divides the population into two compartments:

- **Susceptible (S):** Individuals who have not yet received the information but are potential recipients.
- **Infected (I):** Individuals who have acquired the information and can transmit it to others.

The core assumption of the SI model is that once individuals receive the information, they remain in the infected state indefinitely, continuously spreading the information. This assumption simplifies the model and focuses on the growth of information spread without considering factors like forgetting or loss of interest.

The dynamics of the SI model are governed by the following differential equation:

$$dI/dt = \beta \times (S \times I) / N$$

Where:

- β is the transmission rate of the information.
- **S** is the number of susceptible individuals at time **t**.
- **I** is the number of infected individuals at time **t**.
- **N** is the total population size ($N = S + I$).

This equation models the rate at which susceptible individuals become infected through interactions with infected individuals. Since:

$$dS/dt = -dI/dt$$

The decrease in susceptible is equal to the increase in infected individuals.

Rationale for Using the SI Model

Our research leverages the SI model due to its minimal assumptions and computational efficiency. By focusing on the essential mechanism of information transfer—interactions between informed and uninformed individuals—we can derive a simplified yet powerful explanation of information diffusion. This simplicity allows us to:

- **Analyze Large Networks Efficiently:** The SI model's computational simplicity makes it suitable for simulating information spread in large-scale social networks without significant computational overhead.
- **Identify Influential Nodes:** By modeling the spread dynamics, we can pinpoint key individuals (nodes) whose influence accelerates the dissemination of information across the network.

Comparison with More Complex Models

While the SI model provides a foundational understanding, other models like **SIR (Susceptible-Infected-Recovered)** and **SIS (Susceptible-Infected-Susceptible)** introduce additional compartments to capture more complex behaviors.

SIR Model:

- **Recovered (R):** Individuals who have received the information but are no longer spreading it.
- **Dynamics:**
 - $dS/dt = -\beta \times (S \times I) / N$
 - $dI/dt = \beta \times (S \times I) / N - \gamma \times I$
 - $dR/dt = \gamma \times I$
- **Assumptions:** Introduces a recovery rate γ , representing the rate at which individuals stop spreading information.

SIS Model:

- **Susceptible-Infected-Susceptible:** Individuals can return to the susceptible state after being infected.
- **Dynamics:**
 - $dS/dt = \gamma \times I - \beta \times (S \times I) / N$
 - $dI/dt = \beta \times (S \times I) / N - \gamma \times I$
- **Assumptions:** Assumes a reinfection process where individuals can repeatedly become infected and spread the information.

Limitations of SIR and SIS Models

SIR and SIS models introduce recovery and reinfection dynamics but rely on assumptions that may not always hold, adding unnecessary complexity to the analysis. Estimating recovery rates and assuming reinfection can be impractical and may obscure fundamental insights into information spread.

Preference for the SI Model and Centrality Measures

We prefer the SI model for its simplicity and minimal assumptions, allowing clear focus on identifying influential nodes without confounding factors. By using centrality measures like degree, betweenness, and closeness centrality within the SI framework, we can effectively pinpoint key individuals to optimize information diffusion.

CHAPTER 3: FRONT END, BACKEND AND SYSTEM REQUIREMENT

3.1 Front End

The project will include a dynamic front-end using **React.js** in the future. This will allow users to visualize the diffusion of information in real-time, input parameters, and observe network structures through an intuitive interface.

3.2 Backend

Our backend is currently built using **Python**, **Neo4j**, and **NetworkX** for network analysis and model simulation.

- **Neo4j**: A graph database that helps store and analyze complex network structures. It allows efficient querying to identify influential nodes.
- **NetworkX**: A Python library for network creation and manipulation, used to simulate information spread and calculate centrality measures.

In the future, we will integrate **Flask** or **Django** as our web framework to build a robust and scalable backend, depending on the project's needs. We also plan to introduce **Mesa**, an agent-based modeling framework, to simulate more detailed, human-like behaviors within the network.

3.3 System Requirements

Hardware Requirements:

- **Processor**: Minimum Intel Core i5; Recommended Core i7 or equivalent.
- **Memory**: Minimum 8 GB RAM; Recommended 16 GB RAM for better multitasking.
- **Storage**: 500 GB HDD/SSD minimum; 512 GB SSD recommended for faster processing.
- **Graphics**: Dedicated GPU recommended for smoother data visualization.

Software Requirements:

- **Operating System**: Linux, macOS, or Windows 10/11.
- **Languages**: Python 3.x (backend), JavaScript (front-end).
- **Libraries**: Neo4j, NetworkX, Flask/Django, React.js (front-end), Mesa (for future agent-based modeling).
- **Database**: Neo4j Community/Enterprise Edition.
- **IDE**: VS Code, PyCharm, or any preferred IDE for Python and JavaScript development.

This setup ensures the efficient operation of both the backend and front-end systems, enabling seamless simulations of information diffusion and network analysis

Chapter 4: Methodology

4.1 Topic of the Work:

In this project, we aim to study the maximization of information spread in large social networks. By using the SocioPatterns dataset, we analyze communication patterns between individuals, model these interactions as a graph, and simulate the spread of information using an SI model. This project leverages Neo4j for graph representation and Messa library in Python for modeling the spread of information over time.

4.2 System Design / Architecture

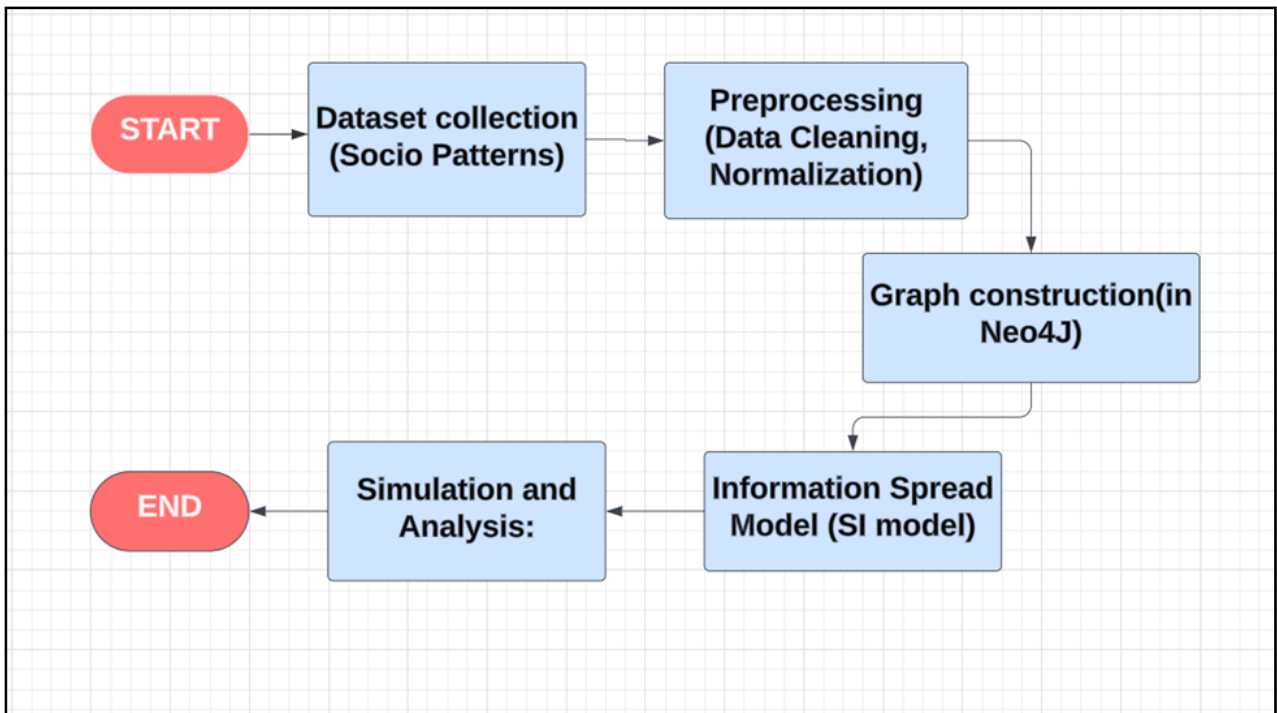


Figure 1: Architecture Diagram

4.2.1 Graph Formation

For this project, we utilized the open-source social network datasets from **SocioPatterns** (<http://www.sociopatterns.org/datasets/>) and **SNAP (Stanford Network Analysis Project)** (<https://snap.stanford.edu/data/index.html>), both of which provide detailed records of interactions between individuals. Each interaction in these datasets is recorded as a pair of nodes (Node 1 and Node 2) along with a corresponding frequency or weight that indicates the strength or frequency of communication between the individuals.

The dataset structure is given below:

	A	B	C	D	E	F
1	SL NO	SRC	DEST	FREQ	NormFreq	
2	0	15	17	13	0.000659479	
3	1	15	21	16	0.000811666	
4	2	15	48	16	0.000811666	
5	3	15	50	23	0.00116677	
6	4	15	63	19	0.000963853	
7	5	15	66	12	0.000608749	
8	6	15	80	22	0.001116041	
9	7	15	102	15	0.000760937	
10	8	15	105	13	0.000659479	
11	9	15	116	18	0.000913124	
12	10	15	118	26	0.001318957	
13	11	15	119	15	0.000760937	
14	12	15	122	17	0.000862395	
15	13	15	134	23	0.00116677	
16	14	15	150	23	0.00116677	
17	15	15	153	20	0.001014582	
18	16	15	154	21	0.001065311	
19	17	15	179	23	0.00116677	
20	18	15	209	15	0.000760937	
21	19	15	240	12	0.000608749	
22	20	15	271	21	0.001065311	
23	21	15	275	19	0.000963853	
24	22	15	285	22	0.001116041	
25	23	15	335	26	0.001318957	

Figure 2: Dataset Preview

Step 1: Data Preprocessing

Before normalizing the data, we performed a preprocessing step to handle the large and complex nature of the dataset. This involved cleaning the data by removing any missing values, duplicates, or erroneous records that could skew the results. We also aggregated interactions between the same pair of nodes to ensure each node pair was represented by a single edge with a cumulative frequency. Additionally, we filtered out low-frequency interactions that might not have a significant impact on the diffusion process, thereby reducing noise and improving the overall efficiency of the model.

Step 2: Data Normalization

Since raw frequencies don't directly indicate the strength of relationships, we normalize these frequencies. Normalization converts the raw values into probabilities that represent the relative communication intensity between two nodes. This ensures that our model reflects realistic interaction probabilities, where stronger connections are more likely to spread information.

Step 3: Graph Representation

We created two versions of the social network graph using different tools:

- **Neo4j:** We use Neo4j to create a directed graph. Here, nodes represent individuals, and edges (or connections between nodes) represent the normalized communication frequencies. Neo4j's graph database makes it easy to query and analyze these connections, helping us to identify key influencers in the network.
- **NetworkX (Python):** In parallel, we replicate the same graph structure in Python using the NetworkX library. NetworkX provides powerful algorithms for analyzing graph properties, such as centrality measures, and simulates how information spreads through the network.

Step 4: Future Implementation with Mesa

In the next phase of the project, we plan to integrate the **Mesa** library. Mesa will allow us to add agent-based modeling, where each node acts as an agent with its own behavior, mimicking the complexity of real-world social interactions. This will add a more dynamic element to our simulations and improve our understanding of how different agents (people) react to or spread information.

4.2.2 Information Spread Modeling

The core of our simulation is based on the **SI (Susceptible-Infected)** model. This epidemiological model is often used to simulate how diseases spread, but in our project, we repurpose it to simulate how information spreads through social networks.

- **Susceptible (S):** These are individuals in the network who have not yet received the information (or message).
- **Infected (I):** These are individuals who have already been informed.

4.3 Working Principle

Step 1: Initialization

The process starts by selecting a percentage of nodes that are "infected" at the outset. These initial nodes act as the starting points for information diffusion, which can represent anything from a viral message on social media to the spread of important news.

Step 2: Propagation of Information

As the simulation progresses, infected nodes pass on the information to their susceptible neighbors, following the probabilities set by the normalized edge weights. The stronger the connection between two nodes, the higher the chance that the information will spread.

Step 3: Role of Key Nodes

During the simulation, certain nodes emerge as particularly important for spreading the information quickly. These are typically nodes with high centrality measures, meaning they have a large number of connections or act as critical bridges between different parts of the network. Identifying these key nodes helps in understanding how information can be spread most effectively.

Step 4: Stopping Criteria

The simulation runs until every node in the network has been informed. This final state is often referred to as "complete infection" in the context of the SI model. The time it takes to reach this point, as well as the pathways the information followed, provide insights into the efficiency of the diffusion process.

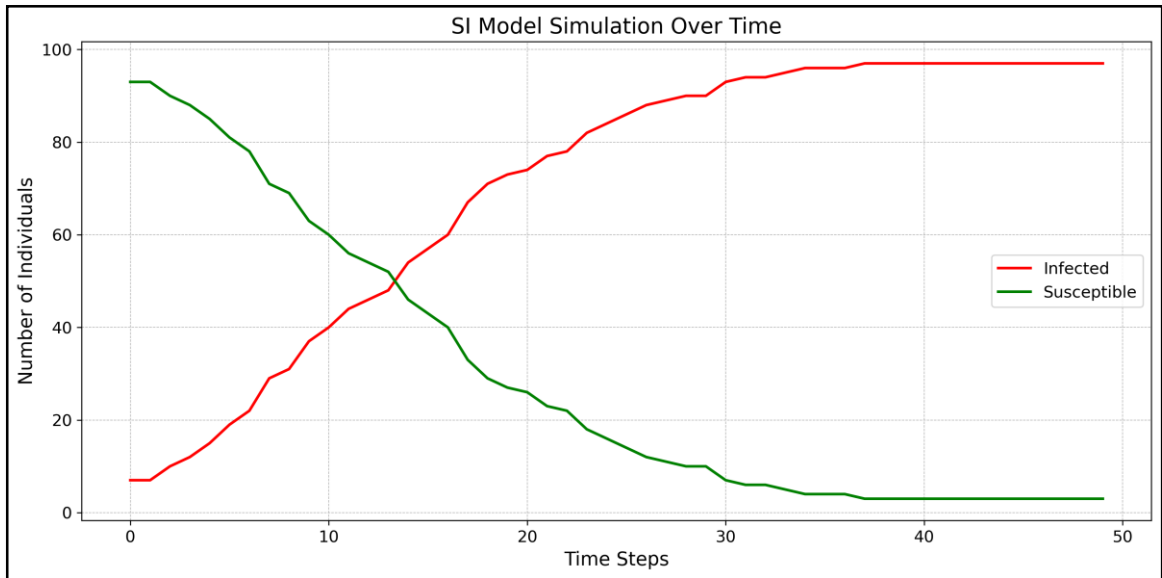
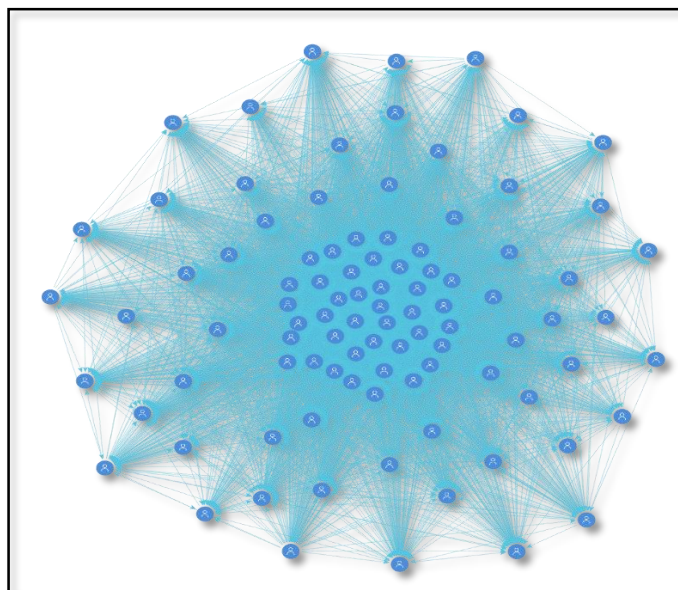


Figure 3: Basic SI Model Visualization using NetworkX library

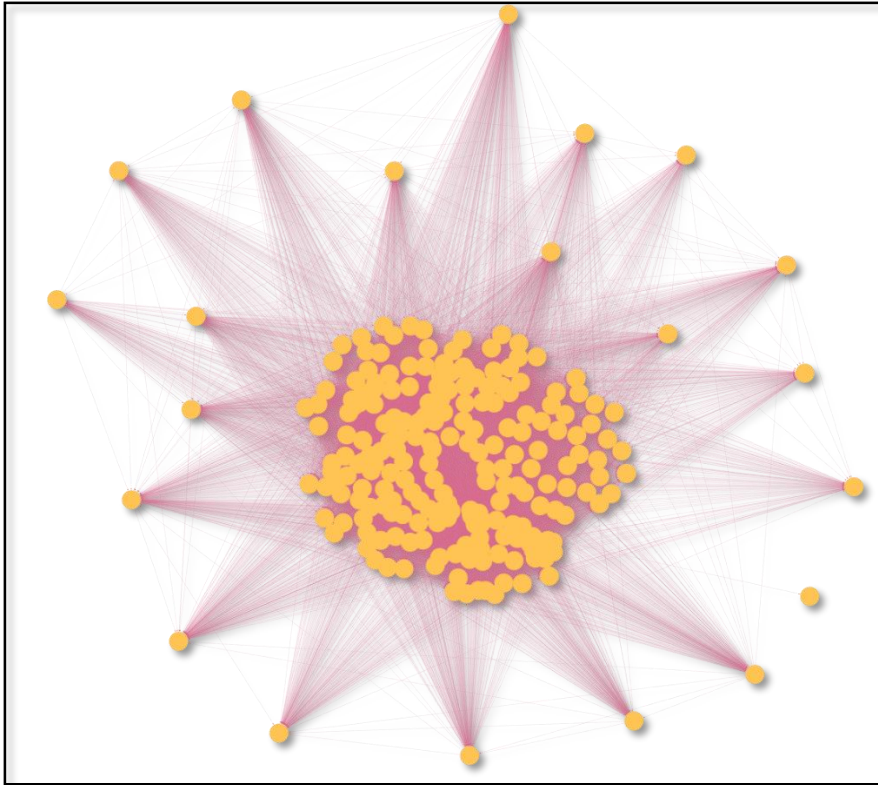
4.4 Results and Visualization

As part of the results, we present several screenshots from our **Neo4j graph database**, showcasing **modeled instances of real-world social contact networks**. These networks have been created using three different datasets from the **SocioPatterns** and **SNAP** repositories, representing interactions between individuals in various environments. Each node in the graph represents an individual, while the edges (connections) represent the frequency or intensity of communication between them.

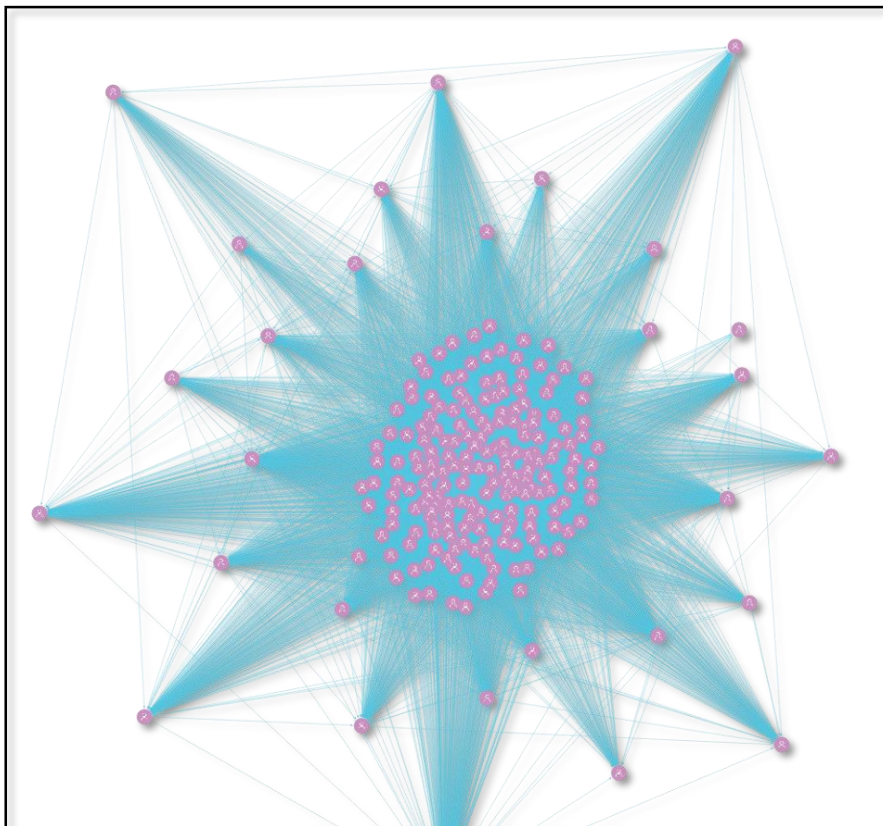
The visualizations demonstrate how we have structured these real-world datasets into directed graphs, where the nodes and edges help us simulate the diffusion of information. The different graphs highlight variations in network topology, such as dense clusters of nodes that indicate close-knit groups' with frequent interactions and sparse regions representing less connected individuals.



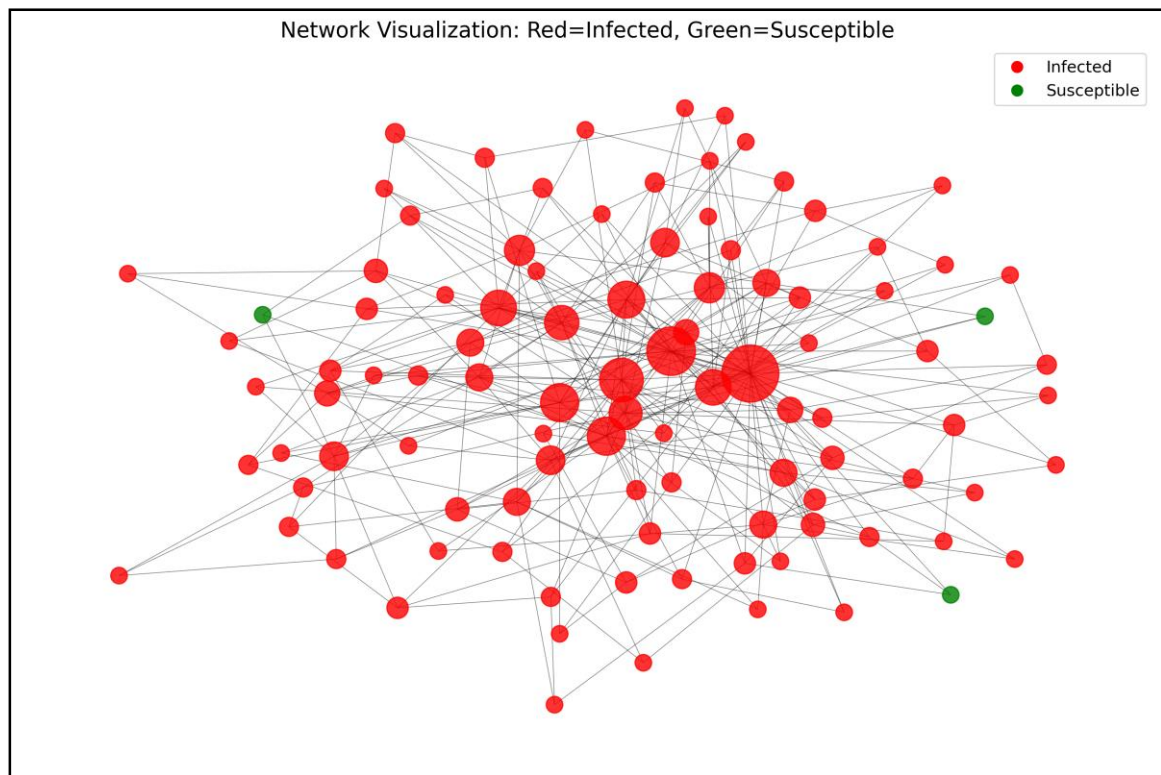
Social contact data from the InVS13 Dataset.



Social contact data from the LyonSchool Dataset.



Social contact data from the InVS15 Dataset.



Visualization using matplotlib, networkx in Python

4.5 INDIVIDUAL CONTRIBUTION BY MEMBERS

4.5.1 DEVASHISH DHAULAKHANDI - 21BCE10076

ROLE: Backend – Equation Modeling, Research, Neo4j Simulation

Devashish's role in the project centered on **equation modeling**, conducting extensive **research** to establish a solid theoretical foundation for the project's simulations. He was responsible for reviewing existing models, particularly the SI model and its applications to social networks, and adapting these for our implementation. By analyzing various mathematical models, he ensured that the diffusion of information was accurately captured based on real-world interaction patterns. Devashish researched key concepts, such as centrality measures and network dynamics, to enhance the understanding of how information flows through social networks. This foundational work was crucial for optimizing the project's simulation strategies.

In addition to research, Devashish was responsible for **graph modeling** and simulations using **Neo4j**, a graph database. He designed and modeled the network structures by converting raw datasets into graph structures that represented nodes (individuals) and edges (interactions). Neo4j allowed him to visualize and simulate large-scale social networks, where he could run various diffusion models and analyze the role of influencers within these networks. His contribution ensured that the project had a robust graph architecture capable of handling complex simulations and delivering meaningful insights into network behavior.

4.5.2 ASWIN P KUMAR - 21BCE11072

ROLE: Backend – Python NetworkX Implementation, Research

Aswin took the lead on the **Python backend development** and **NetworkX implementation**, working closely with the data and network simulation aspects of the project. His role began with the crucial task of processing large datasets from SocioPatterns and SNAP, where he handled **dataset preprocessing** to ensure data consistency and accuracy. Aswin was responsible for converting these datasets into graph structures within NetworkX, a Python library designed for network analysis. He conducted simulations using the SI model, calculating centrality measures to identify key influencers within the network. This Python-based approach provided flexibility, allowing the team to explore various configurations and perform computational analyses efficiently.

Aswin also contributed significantly to the **research** component of the project, focusing on the practical application of network theory in information diffusion. He explored how different network topologies and structures impact the speed and reach of information spread. His research helped refine the parameters used in the simulation, ensuring the model accurately reflected real-world social networks. Additionally, Aswin investigated the role of social dynamics in digital and physical spaces, which informed the design of the agent-based simulations that would be incorporated in future phases of the project.

4.5.3. BRAJ KISHOR SHARMA- 21BCE11078

ROLE: Frontend – React Development, Research

Braj played a key role in developing the **frontend** of the project, working extensively with **React.js** to build an intuitive user interface. His task was to create a dynamic and responsive platform that allows users to interact with the backend simulations easily. Braj's focus was on ensuring that users could input parameters, observe the spread of information through visualizations, and modify settings to see the results of different simulations in real-time. His development work included designing components that displayed the network structures and enabled users to analyze key metrics such as the number of informed nodes over time.

Braj also contributed to the **research** aspect, focusing on how user interactions can be better designed to represent complex simulation data. He explored best practices in user experience (UX) and front-end development for data-driven applications, helping the team understand how to present network simulation results effectively. His research led to enhancements in the interface, making it more accessible and allowing users to easily grasp the implications of the simulations. This blend of development and research helped ensure the project's visual and functional success.

4.5.4. ARIJIT MODAK- 21BCE11159

ROLE: Frontend – React Development, Research

Arijit worked closely with Braj on the frontend development using React.js, contributing to the overall user interface design. His role involved creating key components of the platform, such as interactive visualizations, input forms for simulation parameters, and real-time updates on information diffusion across the network. Arijit focused on making the interface intuitive and user-friendly, ensuring that users could easily interact with the complex backend simulations. He also implemented key features that allowed users to dynamically modify simulation settings and instantly see the results, contributing to the platform's interactivity.

In addition to development, Arijit conducted research on how to effectively present complex network simulations in a clear and understandable way. He reviewed existing visualization techniques in social network analysis and incorporated best practices for data visualization into the project. His research helped the team design a frontend that not only looked appealing but also conveyed the results of the simulations in a meaningful and user-friendly manner. Arijit's contributions in both development and research ensured that the frontend provided a seamless and insightful experience for users.

4.5.5. AKSHANT UJJWAL – 21BCE11650

ROLE: Documentation, Research

Akshant was responsible for managing the project's documentation, ensuring that all aspects of the project were thoroughly recorded. He created detailed technical documentation for the backend and frontend components, describing the system architecture, graph modeling, and data processing steps. Akshant's work also included documenting the simulations, explaining how different parameters were used in the SI model and how the results were analyzed. His detailed descriptions provided clarity on how to reproduce the results, making the project more accessible to future developers or researchers.

Apart from documentation, Akshant contributed to the research by reviewing literature on social network analysis and information diffusion. His research provided the team with additional context on how information spread models could be applied in real-world scenarios, such as marketing or crisis communication. Akshant also ensured that the research findings were reflected in the project's documentation, helping bridge the gap between theoretical concepts and practical implementation. His contributions were critical in maintaining a high level of organization and coherence throughout the project.

CHAPTER 5: CONCLUSION

In conclusion, our project successfully modeled **information diffusion** in real-world social networks using the **SI (Susceptible-Infected) model**, demonstrating how information propagates through connected individuals in a network. By leveraging **Neo4j** for graph-based visualization and **NetworkX** for Python-based network analysis, we effectively simulated the dynamics of information spread. The open-source datasets from **SocioPatterns** and **SNAP** allowed us to work with real-world social interactions, which made our simulations more realistic and reflective of actual human behavior. This enabled us to identify **key influencers** within these networks—nodes with high centrality that play pivotal roles in the spread of information—and to analyze how different network structures influence the speed and reach of information propagation.

The final outcome of the project highlights the critical role of understanding **social network dynamics** in various practical applications. Our results indicate that nodes with higher centrality measures, such as degree or betweenness centrality, significantly accelerate information diffusion. This finding has major implications for domains like **digital marketing**, where targeting these key nodes can maximize the reach of campaigns, and **public health**, where efficiently disseminating critical information can control the spread of misinformation or promote public awareness. Our **Neo4j visualizations** provided an intuitive way to observe these patterns, helping us better understand the flow of information in different types of networks. These insights are valuable in fields such as **misinformation control**, **crisis communication**, and **viral marketing**, where the timely and effective spread of information is crucial.

5.1 Future Scope

Looking ahead, the project lays the groundwork for several **future developments**. One key area of expansion is the integration of more **complex models** like the **SIR (Susceptible-Infected-Recovered)** and **SIS (Susceptible-Infected-Susceptible)** models, which incorporate recovery and reinfection dynamics. These models would allow us to simulate real-world scenarios more accurately, where individuals can lose interest in a piece of information (recovery) or be exposed again (reinfection). Additionally, the next phase will involve implementing **agent-based modeling** using the **Mesa** library. This will enable us to model more nuanced, individual-level behaviors, where each node (or agent) acts independently, reflecting human decision-making and interaction patterns.

Moreover, we plan to explore the impact of **external factors** such as media influence, societal trends, and online versus offline interactions on information diffusion. This could provide a more comprehensive understanding of how information spreads in today's multi-channel communication landscape. Integrating **real-time data** and improving the scalability of our simulations will also allow us to study larger, more diverse networks in the future. This expansion can contribute to a wide range of fields, including **social media analytics**, **public policy formulation**, and **consumer behavior analysis**, where understanding information flow is critical for decision-making and strategy development.

CHAPTER 6: REFERENCES AND PUBLICATIONS

1. **Wikipedia. (n.d.). Compartmental Models in Epidemiology.**

Link: https://en.wikipedia.org/wiki/Compartmental_models_in_epidemiology

Provides an overview of basic epidemiological models like SIR, SIS, and SEIR, which serve as the foundation for the project's simulations.

2. **ArXiv. (n.d.). Influence Maximization in Social Networks.**

Link: <https://arxiv.org/abs/>

This paper offers detailed explanations of algorithms for influence propagation in networks, relevant to identifying key influencers for information diffusion.

3. **EON: Exponential Random Graph Models.**

Link: <https://link.springer.com/book/10.1007/978-3-319-50806-1>

This book offers foundational knowledge on disease spread modeling and highlights the limitations of traditional approaches, relevant to improving our modeling framework.

4. **Barabási, A.-L. (2016). Network Science. Cambridge University Press.**

Link: <https://networksciencebook.com/>

A comprehensive source of knowledge on network science, covering models and simulations, including popular differential equation-based approaches used in network modeling.